

Black-box Active Learning for Discovering New Explanations



Derivative-Free Optimization Symposium

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Context

- *Multi-agent systems & simulation*
- *Explainable AI for complex systems*
- *Socio-environmental digital twins*

600 lab members
Yearly budget 5.1M€ (2023)



Membre de l'Université Toulouse
Capitole

Created in 1229
3 campus
750 Teaching staff
740 administrative staff
21000 students
151 national diplomas

ANR-JCJC MIMICO: MIMIcking the COmplexity of agent-based models

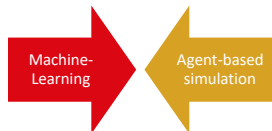
■ Team



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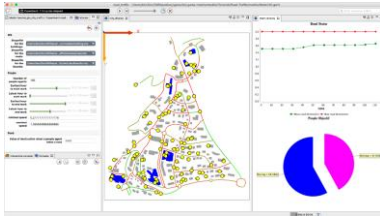
Complex Systems Simulations

■ Agent-based simulations

- Useful to study **complex socio-environmental** challenges (climate change, water scarcity, urbanization, pollution, energy transition, sustainability...)
- Simulation models can be abstracted as a program that transforms a set of **inputs** into a set of **outputs**.
- **Calibration/Exploration/Optimization:** goal-oriented and user-defined automated exploration of a model's outcomes to study a given emerging and decentralized phenomenon



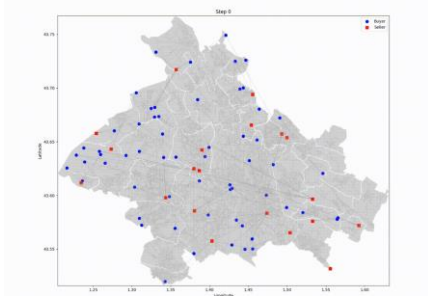
<https://gama-platform.org/>



From Agent based simulations to emergent behaviors

- Industrial Symbiosis model - MASTIO

- ⚙ Initial **random** placement, **RL algo.**, decreasing temp.

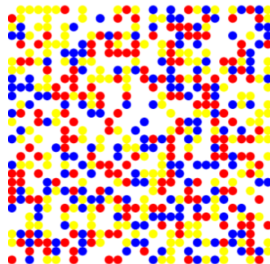


- ⚙ 5 cont.: product scarcity, firm density, transport. cost, disposal cost, cluster spread



- Schelling segregation model

- ⚙ Initial **random** placement, **random** move if unsatisfied.

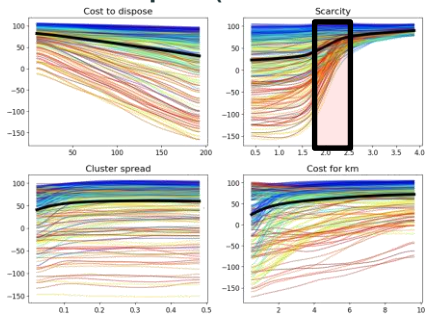


- ⚙ 3 integers: grid size, number of socio-types, agent perception distance & 2 cont.: intolerance of uninhabitant and population density.

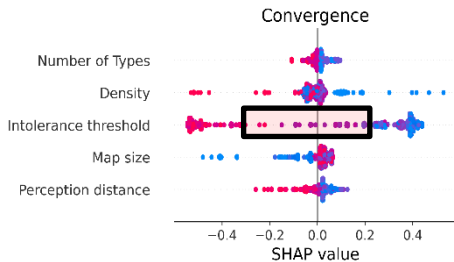
<https://mobiliscope.cnrs.fr/en/geoviz/montreal>

Motivation: Tipping Points in Dual Spaces

Parameter Space (MASTIO ICE/PDP)



Explanatory Space (Schelling SHAP)



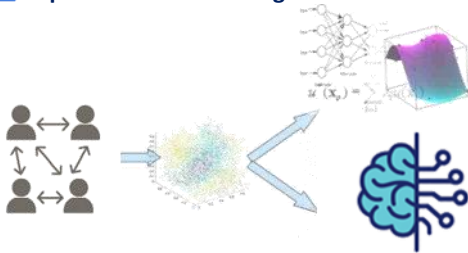
- Tipping point: Prediction collapses/diverges at specific parameter thresholds.
- Transient phases may exhibit scale-free (power-law) dynamics.
- Internal logic shift: Appearance of isolated "islands" where features suddenly dominate convergence.

RQ: Can hunting for explanatory novelty help find physical tipping points?

Limits of Complex Systems Simulations

- **High computation costs:** large numbers of agents inter. **over time and space.** *High-dimensional mapping via LHS is intractable.*
- **Opacity of outcomes:** intricate interactions among agents often lead to results that are **hard to trace back** to specific inputs or rules. *Not easily decomposable.*
- **Black-Box:** No gradients to guide the search towards interesting behaviors.
- **Needle in a Haystack:** Traditional Active Learning minimizes *prediction uncertainty*, wasting simulations in noisy but homogeneous regions while missing critical tipping points.

Explainable AI for Surrogate models



- **Mimic** a complex system, black box $\Rightarrow$ ease calibration, scenarios exploration, and optimization
- Types of models used $\Rightarrow$ Models are **complementary: complexity or explicability** based on **machine learning**

Pipeline & Innovations

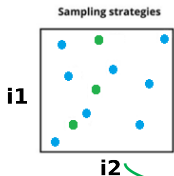
Active Learning and localities

Local explainability

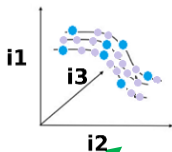
Q2: What does SM tell us about the parameter space?

Exploration + Calibration

Initial
dataset



Surrogate model



Refined
dataset

Q1: How to efficiently train SM from a few samples?

Initial + Refined datasets

Global explainability

Q3: How does the SM inform us about the model dynamics?

Analysis + XAI

Augmented
dataset



The Core Concept: Explanatory Diversity

The Shift: Stop hunting for *prediction uncertainty*; start hunting for *explanatory novelty*.

Instead of looking only at the prediction y , we extract feature contribution profiles using Explainable AI (SHAP):

$$S(x) = \text{SHAP}(f, x)$$

The Goal: Explicitly sample regions where the internal logic (the explanation) of the model fundamentally changes.



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Thank you for your attention!



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The Active Learning Pipeline Architecture

How does surrogate models inform us locally or instance-wise?

What can surrogate models tell about the overall model dynamics?

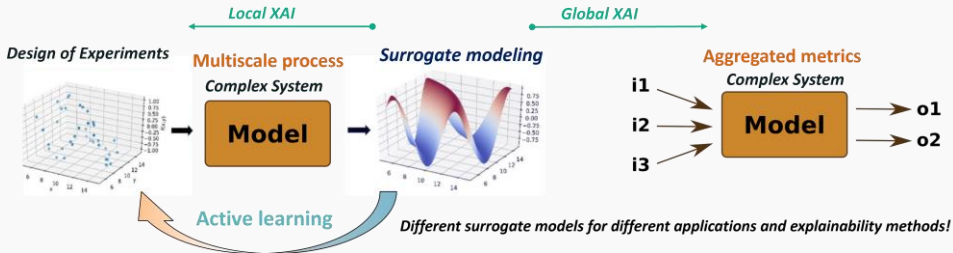


Figure 1: Flowchart of the Proposed Active Learning Framework

Algorithmic Innovations (I): Fast Cosine Proxy

From Slow KDE to Fast Cosine Proxy:

- The initial theory relied on high-dimensional Kernel Density Estimation (KDE) on SHAP values:

$$p_{SHAP}(S(x)) = \frac{1}{N} \sum_{i=1}^N K_h(S(x) - S(x_i))$$

- The Upgrade:** KDE suffers from the curse of dimensionality and is computationally prohibitive.
- The Solution:** We implemented a **Fast Density Proxy via Cosine Distance** (Nearest Neighbors) in SHAP space. High cosine distance translates directly to high explanatory novelty.

Algorithmic Innovations (II): Bi-Objective Optimization

Framing Dynamic-US as a Bi-Objective Problem:

Rather than a static heuristic, the acquisition strategy sweeps a Pareto front to simultaneously:

1. **Maximize Physical Exploration:** Predictive variance in input space (σ_{GP}^2).
2. **Maximize Explanatory Exploitation:** Novelty via Cosine distance in SHAP space (EI_{MFK}).

$$a_{Dynamic}(x) = (1 - \alpha^{(t)}) \cdot \tilde{\sigma}_{GP}^2(x) + \alpha^{(t)} \cdot \tilde{EI}_{MFK}(S(x))$$

- Over iterations, $\alpha^{(t)}$ shifts dynamically from 0.0 $\rightarrow$ 1.0.
- **Result:** It maps the macroscopic behavior first, then pivots to isolate the complex **tipping points**.

Results & Analysis

Hunting Tipping Points in PCA Space

- Traditional methods (LHS, Space-US) sample densely in the homogeneous bulk.
- Our method (**SHAP-US** and **Dynamic-US**) efficiently populates the boundaries and critical tipping points.

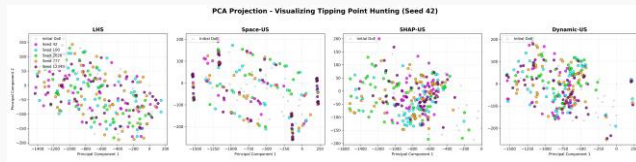
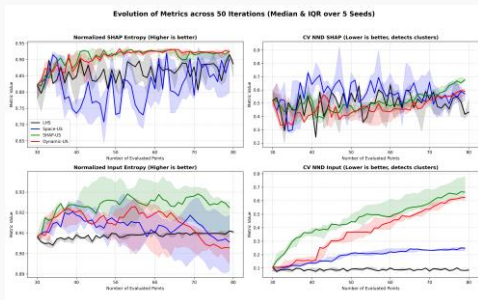


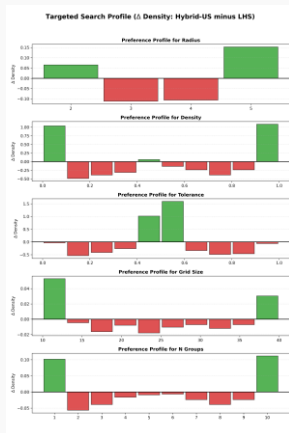
Figure 2: Exploration of the Cireco Model (PCA space)

Convergence and Stability over Time



- **Dynamic-US (Red)** matches the high explanatory diversity (SHAP entropy) of SHAP-US.
- Crucially, it avoids the catastrophic crash in physical input entropy seen in pure SHAP-US.
- It converges reliably without pathological spatial clustering.

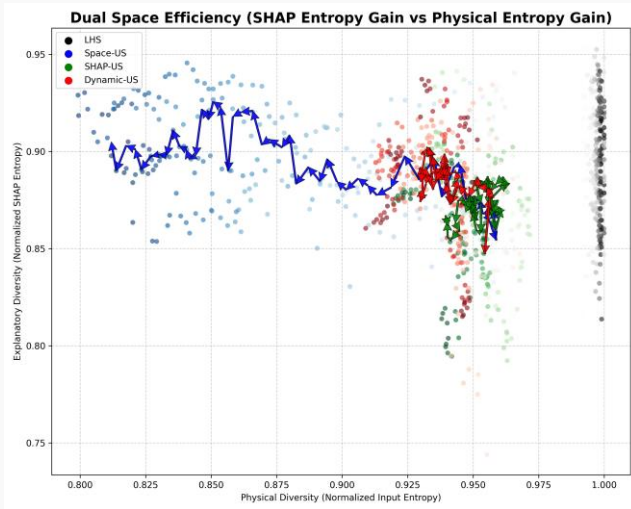
Under the Hood: How Dynamic-US Samples Differently



- Shows the Delta Density (**Dynamic-US minus LHS**).
- **Green:** Active learner intentionally over-sampled this region.
- **Red:** Actively ignored.
- Validates that Dynamic-US aggressively targets specific narrow bands (e.g., Tolerance around 0.5) where the tipping points live.

Dual Space Efficiency

- **Physical Diversity vs Explanatory Diversity.**
- Dynamic-US achieves a rapid gain in explanatory entropy, proving it discovers diverse causal mechanisms faster.
- Background dots represent 250 iterations across 5 seeds.



Conclusion

Contributions:

- A model-agnostic, multi-fidelity XAI loop that achieves deep structural understanding rather than just raw predictive accuracy.
- Demonstrates that optimizing for explanatory diversity uncovers tipping points exponentially faster than classic DoE or SUR.
- Validated on robust Agent-Based Models (Cireco, Schelling).

Scaling the Optimization and Dimensionality:

- **Integrated Variance Reduction (IMSE):** Replacing pointwise GP variance (σ^2_{GP}) with Integrated Mean Squared Error to solve the extrapolation artifacts at the boundaries of the hypercube.
- **Active Subspace Projection:** Mathematically linking SHAP tracking to the dynamic rotation of the Active Subspace. This provides a theoretical pathway to scale to $M > 15$ dimensions by reducing ExactSHAP overhead.
- **Native Simulator Integration:** Transitioning the surrogate methodology from the MLP proxy directly to the computationally intensive ABM cores.

Advanced Surrogate & Acquisition Optimization:

- **Multi-Objective DFO (EHVI):** Replacing linear scalarization with Expected Hypervolume Improvement to capture non-convex regions of the Pareto front.
- **Trust-Region Bayesian Optimization (TuRBO):** Using dynamically sized local trust regions to handle the inherently non-stationary landscapes of SHAP novelty.
- **Global Solvers (CMA-ES):** Upgrading the acquisition optimizer from L-BFGS-B to Covariance Matrix Adaptation Evolution Strategy to avoid local minima traps in the multimodal novelty surface.